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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BCA (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Elective (DSE) - MG3DSEBCA200 - Basics of Data Analytics Using Spreadsheet

Duration: 1 Hours **Maximum Marks: 35**

Part A
Short Answer Questions (Answer any 7 questions, Each carries 5 marks)

1. Discuss the usage and importance of data analytics.

Data analytics involves examining raw data to draw conclusions about the information, identify patterns, and support decision-making. It is applied across various fields, including business, healthcare, and finance.
Importance: It provides the foundation for effective decision-making , helps businesses optimize operations, uncovers hidden trends, aids in predictive forecasting, and is crucial for real-world problem-solving and business analytics.

2. Explain the application of data analytics in the healthcare sector. How does it help in improving patient care and operational efficiency?

Application: In healthcare, data analytics is used for tasks like **predictive diagnostics**, managing chronic diseases, and **optimizing resource allocation**.

- **Improving Patient Care:** It analyzes patient data to personalize treatment, predict disease outbreaks, and improve the **quality of care**.
- **Operational Efficiency:** It helps optimize hospital staffing, manage equipment, track resource utilization, and can lead to a reduction in wait times and lower administrative costs.

3. Estimate the proportion of unstructured data in real-world analytics compared to structured data.

Estimation: The majority of data generated today is considered **unstructured** (often estimated to be 80-90%), with **structured data** making up a much smaller proportion (Types of data Analytics).

- **Reason:** The rapid growth of data from sources like social media, emails, images, and sensor data (all of which are unstructured) contributes significantly more volume compared to the relatively fixed volume of traditional data found in databases (structured data).

4. Discuss applications of predictive analytics.

Predictive analytics uses historical data and statistical models to forecast future outcomes or trends. It identifies patterns and probabilities to make educated guesses about what might happen.

- **Applications:**
    - **Finance:** Fraud detection and credit risk assessment.
    - **Marketing:** Predicting customer churn and creating personalized product recommendations.
    - **Operations:** Inventory forecasting and predictive maintenance (to anticipate equipment failure).

5. Demonstrate how data anonymization can protect sensitive information during analysis?

Data anonymization is an **ethical consideration in data analytics** used to protect privacy by ensuring individuals cannot be identified from the data.
Techniques:
Masking/Encryption: Replacing sensitive identifying fields (like names or IDs) with fake or encrypted values.
Generalization: Replacing specific values with a broader category (e.g., replacing a specific age with an age range like '20-30').
Swapping: Exchanging attributes between different records, which maintains the overall statistical properties but breaks the link to specific individuals.
Protection: By removing or obscuring personally identifiable information (PII), anonymization allows for data to be used for manipulation and analysis without violating privacy regulations.

6. Illustrate the process of removing noisy data using binning techniques.

Binning is a technique used in Data Cleaning and Transformation to smooth noisy data (random errors or variance).

Steps:
1. Sorting: The data values are first sorted in ascending order.
2. Partitioning (Binning): The sorted data is divided into equal-sized partitions or "bins."
3. Smoothing: The values within each bin are replaced by a representative value:
    a. Smoothing by Mean (Averaging): Each value in the bin is replaced by the mean (average) of the bin's values.
    b. Smoothing by Median: Each value is replaced by the median (middle) value of the bin.
    c. Smoothing by Boundaries: Each value is replaced by the closest boundary value of the bin (minimum or maximum value of the bin).

7. Explain the major financial applications of data analytics, such as fraud detection, credit risk assessment, and investment forecasting.

These are industry-specific applications (finance) that utilize data analytics.
- **Fraud Detection:** Analyzes transaction data in real-time using algorithms to identify unusual patterns that deviate from normal user behavior, automatically flagging potential fraud.
- **Credit Risk Assessment:** Uses predictive models to analyze a borrower's financial history, debt levels, and credit score to accurately forecast the probability of loan default.
- **Investment Forecasting:** Employs advanced techniques like time-series analysis and machine learning to predict stock price movements, market trends, and asset volatility, guiding strategic investment decisions.

8. Describe importance of Data Transformation in data analysis.

Data transformation is the process of converting data from one format or structure to another, often involving steps like Data Cleaning.

Importance:

- Compatibility: Ensures the data is in the correct format for the specific analytical tools or algorithms being used.
- Data Quality: Improves quality by reducing inconsistencies, redundancies, and improving overall accuracy.
- Normalization/Scaling: Ensures all variables are on a similar scale, preventing features with larger values from dominating the analysis.
- Feature Engineering: Creates new, more insightful variables (features) from the existing data, which significantly improves the performance of predictive models.

9. Explain the significance of outliers.

Outliers are observations that lie an abnormal distance from other values in a random sample from a population (Handling Missing Data and Outliers).

Significance:

- Data Errors: Often, they indicate genuine data entry errors or measurement mistakes that must be cleaned or removed for accurate analysis.
- Valuable Insight: Sometimes, outliers represent rare but significant events (e.g., a major system failure, a highly profitable anomaly, or fraud) that are critical to the analysis.
- Model Distortion: If left unhandled, outliers can severely skew statistical results (like the mean and standard deviation) and reduce the reliability and accuracy of analytical models.

10. Show with examples how structured, semi-structured, and unstructured formats appear in everyday life.

- Structured Data: Highly organized data that fits into a fixed schema (rows and columns).
  - *Example:* A Spreadsheet or database table containing employee records with fixed fields for Employee ID, Name, and Date of Hire.
- Semi-structured Data: Data that has organizational tags or markers but does not conform to a formal table structure.
  - *Example:* An email (has structured "To," "From," and "Date" fields, but the body content is free-form) or a JSON/XML file.
- Unstructured Data: Data with no predefined structure, often text or media based.
  - *Example:* The full body text of a news article, a photograph or video file, or the content of a social media post.

(Only short descriptions were included; **detailed descriptions** are needed for **all** the questions.)

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